Introduction to Software Engineering

Project Idea & Team Contract

The student team is required to complete the **Project Idea & Team Contract** documentation for the assigned course project, following the attached template.



Software Engineering Department
Faculty of Information and Technology
University of Science

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Project Idea & Team Contract

1 Team Information

Item	Value
Team ID	T04
Team name	Matcha kem dẻo
Team lead	Trần Anh Khoa
Communication channel	Discord and Google Meet

Student ID	Full name	Email	Main role	Backup role
24127424	Trần Anh Khoa	takhoa2416@clc.fitus.edu.vn	*<Team lead / PM>*	Nguyễn Trần Phương Thúy
24127248	Nguyễn Trần Phương Thúy	ntpthuy2436@clc.fitus.edu.vn	*<Business analyst>*	Hồng Gia Bảo
24127328	Hồng Gia Bảo	hgbao2428@clc.fitus.edu.vn	*<Designer>*	Lê Nho Duy Phước
24127508	Lê Nho Duy Phước	lndphuoc2424@clc.fitus.edu.vn	*<Developer>*	Trần Anh Khoa
24127424	Trần Anh Khoa	takhoa2416@clc.fitus.edu.vn	*<Tester>*	Nguyễn Trần Phương Thúy

Roles may change later; confirm the final assignment in PA1. Every role needs one main person and one backup.

2 Project Idea Overview

2.1 Project Name and One-Sentence Pitch

ImpactHub is a web application that combines a community forum with fundraising campaigns, real-time donation progress tracking, charity marketplace, and volunteer coordination to connect people in need with those willing to help.

2.2 Problem and Motivation

Currently, stories about people in need are often scattered across social media platforms, making it difficult to verify information and provide support effectively. While social media enables community discussions, it lacks integrated fundraising and donation tracking tools. Meanwhile, crowdfunding platforms mainly focus on collecting donations rather than coordinating volunteers and aid distribution.

ImpactHub aims to address these issues by combining a **community forum, fundraising platform, and charity marketplace** in one system. Users can share stories, create verified fundraising campaigns, track donation progress through visual progress bars. The platform also supports volunteer coordination and allows users to sell products while contributing a portion of their sales to charitable campaigns.

2.3 Target Users

User group	What they need from the system
Donors	Share stories about people in need, interact with forum posts, discover fundraising campaigns, make donations, and track fundraising progress.

Campaign organizers	Create and manage fundraising campaigns, set donation goals, monitor fundraising progress, report expenses, and organize volunteer activities.
Volunteers	Find volunteer opportunities, register for tasks, and track assigned activities to support people in need.
Administrators	Moderate forum posts, verify cases, approve fundraising campaigns, review financial reports, and manage users.

2.4 Main Features

#	Feature	Short description	Must-have?
1	User authentication and role management	Allow users to sign up, log in, edit profiles, and use features based on their roles.	Yes
2	Community forum	Allow users to post stories, upload images, comment, react, share, and search for posts.	Yes
3	Case and role verification	Allow admins to check cases, verify organizers, and approve fundraising requests.	Yes
4	Fundraising campaign management	Allow organizers to create, edit, and manage campaigns with donation goals.	Yes
5	Donation tracking	Allow users to make simulated donations and view fundraising progress using progress bars.	Yes
6	Volunteer and task management	Allow organizers to create tasks and volunteers to join and update their progress.	No
7	Charity marketplace	Allow users to sell, browse, and buy products, with part of each sale donated to a campaign.	No

2.5 Similar Product

Category	Similar Product	Main features
Social media	Threads	Sharing posts, commenting, and community interaction.

Crowdfunding	GoFundMe, Give.asia	Fundraising campaigns, donations, progress tracking, charity marketplace

2.6 Platform and Technology Ideas

Platform: A responsive web application that works on both desktop and mobile browsers. We chose the web because donors usually reach a campaign through a shared link, for example on Facebook or Messenger. They should be able to donate right away without installing an app. A single responsive codebase is also realistic for our team in the three-week coding window.

Proposed technology stack (not final):

Layer	Choice	Reason
Frontend	React (Vite) + Tailwind CSS	Component-based, fast to build responsive layouts, familiar to most members
Backend	Node.js + Express (REST API)	Same language (JavaScript) as the frontend, so all members can work across the stack
Database	PostgreSQL	Relational data (users, campaigns, donations) and transaction support to keep donation totals consistent
Authentication	JWT-based login (email + password)	Simple to implement; Google login is a possible extension
Payment	Stripe or VNPay in sandbox/test mode	Demonstrates the donation flow without handling real money
Image storage	Cloudinary	Stores campaign cover images without managing our own file server

Hosting	Vercel (frontend), Render (backend + database)	Free tiers, easy deployment from GitHub
Tools	GitHub (code, pull requests, CI), Figma (UI design), Postman (API testing)	Supports our work standards in Section 3.5

Notes:

Payments will only run in test mode. Real money transfer, withdrawal to bank accounts and legal verification of campaign owners are out of scope.

The final stack will be confirmed in PA1, once we have checked every member's experience. If most members know another stack better, we may switch.

2.7 Feasibility, Scope and Open Questions

Our main goal is to build a working website where users can create campaigns, make demo donations, track funding progress, and join community discussions.

Out of scope:

1. Real-money transactions or bank withdrawals.
2. A separate mobile application.
3. Advanced AI systems for detecting fraud.
4. Complex volunteer management.
5. Advanced payment and financial management features.
6. Official identity verification and background checks for users and organizers
7. Professional product quality inspection and authentication in the charity marketplace

Main risks:

1. We may have limited time to complete all planned features.
2. Connecting to a payment service's test environment may take more time than expected.
3. Automatically checking forum posts may not always give accurate results.
4. Combining multiple features into one system may be challenging.

5. Ensuring accurate donation tracking and marketplace contribution calculations may be challenging.

To reduce these risks, we will divide the work into smaller tasks, test features regularly, and simplify features.

Questions for the TA:

1. Is our project scope suitable for the course and development timeline?
2. Is using a demo wallet and a payment service's test environment acceptable?
3. Should we simplify or remove any features?

3 Team Contract

3.1 Goals and Expectations

Our team aims to achieve a grade of 10/10 by completing the project requirements.

Each member is expected to spend around 9–10 hours per week working on the project.

3.2 Communication

- ✔ Main communication channel: Discord.
- ✔ Expected response time: Within 12 hours on weekdays.
- ✔ Documents and shared files: Google Drive.
- ✔ Source code: GitHub.
- ✔ Important decisions: Recorded in Google Drive or the team's task board.

3.3 Meetings

- ✔ Weekly meeting with the TA: Online at 20:30 Sunday
- ✔ Internal team meetings: 21:00 Friday
- ✔ Weekly status report deadline: 23:30 Tuesday

All members are expected to attend team meetings.

If a member cannot attend, they should inform the team at least 4 hours in advance. Missed meetings will be recorded in the weekly status report.

3.4 Roles and Responsibilities

- Team Lead/PM: Set up GitHub, GitHub Projects task board, assign tasks, manage deadlines, and track team progress.
- Business Analyst: Analyze requirements, define features, write user stories and use cases, and organize documents.
- Designer: Design system architecture and UI/UX wireframes.
- Developer: Implement frontend and backend features, integrate system components, and manage the database.
- Tester: Write test cases, perform system testing, report bugs, and verify fixes.

3.5 Work Standards

✔ Internal deadline: Each member finishes their part 3 days before a PA deadline

✔ Quality: Every document section and pull request is reviewed by at least one other member

✔ Tools: Every task is on the task board with an owner and due date; code goes through pull requests

3.6 Use of AI Tools

✔ Allowed tools: Any platforms

✔ Allowed uses:

- Brainstorming ideas
- Explaining unfamiliar concepts or technologies
- Suggesting document structure
- Drafting text that we then rewrite
- Generating boilerplate code
- Suggesting test cases
- Debugging support

✔ Not allowed:

- Submitting AI output without reading and editing it
- Using AI to fill in another member's assigned part
- Entering personal data or other members' private information into AI tools

✔ Review of AI output:

- The member who used AI is responsible for checking the output for correctness and for consistency with our project.
- All AI-assisted content goes through the normal review in Section 3.5: at least one other member reviews it before it is merged or submitted.
- AI-generated code must be understood by its author, who must be able to explain it in reviews and demos.

✔ AI Usage Log:

- Each member adds their own entries to the AI Usage Log (Section 4) for every significant AI use, by the team's internal deadline.
- Each member must keep their own log for each PA so that the logs can be retrieved easily for later purpose. Moreover, any changes by an agent must be verified thoroughly and understood by himself/herself.
- If no AI was used in a PA, the log states "No AI tools were used".

3.7 Decision Making

Decisions are made through team discussions and majority voting. In case of a tie, the team leader makes the final decision.

3.8 Handling Conflicts and Missed Work

✔ Step 1: Talk directly with the member to understand the problem and agree on a solution or a new deadline.

✔ Step 2: If the problem continues, discuss it as a team and try to find a fair solution together. Tasks and deadlines may be adjusted if necessary.

✔ Step 3: Only if the team cannot resolve the issue internally, inform the TA and ask for guidance.

We agree to handle conflicts respectfully, communicate openly, and prioritize solving problems within the team before seeking outside help.

3.9 Signatures

By signing, each member agrees to follow this contract.

Student ID	Full name	Signature	Date
24127424	Trần Anh Khoa		14/10/2026
24127248	Nguyễn Trần Phương Thúy		14/10/2026
24127508	Lê Nho Duy Phước		14/10/2026
24127328	Hồng Gia Bảo		14/10/2026

4 AI Usage Log

#	Member	Tool	Task / purpose	How the output was verified and changed
1	Nguyễn Trần Phương Thúy	ChatGPT	Brainstormed project ideas and analyzed similar products.	Reviewed AI suggestions, compared them with existing platforms, and refined the features based on team discussions and project feasibility.
2	Trần Anh Khoa	ChatGPT	Launch project scope, manage project schedule and team contract ideas.	Discussed the suggestions with the team and adjusted them to fit our project.
3	Lê Nho Duy Phước	Claude	Tech Stack and set up rules of use of AI tools	Agent suggested some ideas for our project then we rechecked and updated in order to suit our work.
4	Hồng Gia Bảo	Gemini	Brainstormed UI/UX layout ideas, wireframe structures, and design guidelines for key features.	Reviewed AI layouts against user requirements, adjusted them in Figma, and ensured responsiveness for target devices.